UGN1 — UGN1 TASK 3: PRINCIPAL COMPONENT ANALYSIS

STATISTICAL DATA MINING — D600
PRFA — UGN1

Preparation **Task Overview** Submissions Evaluation Report

COMPETENCIES

4160.1.2 : **Performs Dimensionality Reduction**
The learner performs principal component analysis to make recommendations based on the results.

INTRODUCTION

As a data analyst, you will assess data sources for their relevance to specific research questions throughout your career. In your previous coursework, you have performed data cleaning and exploratory data analysis on your data. You have seen basic trends and patterns and can now start building more sophisticated statistical models. In this task, you will build, test, and use a linear regression model to support the decision-making process.

Prepare the provided cleaned dataset file for linear regression modeling using principal component analysis (PCA). The organizations connected with the given dataset for this task seek to analyze their operations and have collected variables of possible use to support the decision-making processes. You will analyze your chosen dataset using linear regression modeling, create visualizations, and deliver the results of your analysis.

You will complete this performance assessment in the provided WGU virtual lab environment.

Note: The IDE for this assessment is either Anaconda or R Studio, depending on which language you decide to use to complete the task.

REQUIREMENTS

Your submission must be your original work. No more than a combined total of 30% of the submission and no more than a 10% match to any one individual source can be directly quoted or closely paraphrased from sources, even if cited correctly. The similarity report that is provided when you submit your task can be used as a guide.

You must use the rubric to direct the creation of your submission because it provides detailed criteria that will be used to evaluate your work. Each requirement below may be evaluated by more than one rubric aspect. The rubric aspect titles may contain hyperlinks to relevant portions of the course.

*Tasks may **not** be submitted as cloud links, such as links to Google Docs, Google Slides, OneDrive, etc., unless specified in the task requirements. All other submissions must be file types that are uploaded and submitted as attachments (e.g., .docx, .pdf, .ppt).*

A. Create your subgroup and project in GitLab using the provided web link by doing the following:
- Clone the project to the IDE.
- Commit with a message and push when you complete each requirement listed in parts D2 through F4.

  Note: You may commit and push whenever you want to back up your changes, even if a requirement is not yet complete.

- Submit a copy of the GitLab repository URL in the "Comments to Evaluator" section when you submit this assessment.
- Submit a copy of the repository branch history retrieved from your repository, which must include the commit messages and dates.

B. Describe the purpose of this data analysis by doing the following:
1. Propose **one** research question that is relevant to a real-world organizational situation captured in the provided dataset that you will answer using linear regression in the initial model.
2. Define **one** goal of the data analysis. Ensure that your goal is reasonable within the scope of the scenario and is represented in the available data.

C. Explain the reasons for using PCA by doing the following:
1. Explain how PCA can be used to prepare the selected dataset for regression analysis. Include expected outcomes.
2. Summarize **one** assumption of PCA.

D. Summarize the data preparation process for linear regression analysis by doing the following:
1. Identify the continuous dataset variables that you will need to answer the research question proposed in part B1.

   Note: The code for the initial model from Task 1 should be used for Task 3, making sure to exclude any categorical variables.

2. Standardize the continuous dataset variables identified in part D1. Include a copy of the cleaned dataset.
3. Describe the dependent variable and all independent variables from part D1 using descriptive statistics (counts, means, modes, ranges, min/max), including a screenshot of the descriptive statistics output for each of these variables.

E. Perform PCA by doing the following:
1. Determine the matrix of *all* the principal components.
2. Identify the *total* number of principal components (that should be retained), using the elbow rule or the Kaiser rule. Include a screenshot of the scree plot.
3. Identify the variance of *each* of the principal components identified in part E2.
4. Summarize the results of your PCA.

F. Perform the data analysis and report on the results by doing the following:
  1. Split the data into two datasets, with a larger percentage assigned to the training dataset and a smaller percentage assigned to the test dataset. Provide the file(s).

     Note: The datasets should include only those principal components identified in part E2.

  2. Use the training dataset to create and perform a regression model using regression as a statistical method. Optimize the regression model using a process of your selection, including but not limited to, forward stepwise selection, backward stepwise elimination, and recursive selection. Provide a screenshot of the summary of the optimized model or the following extracted model parameters:
     - Adjusted R^2
     - R^2
     - F statistics
     - Probability F statistics
     - coefficient estimates
     - p-value of each independent variable

  3. Give the mean squared error (MSE) of the optimized model used on the training set.
  4. Run the prediction on the test dataset using the optimized regression model from part F2 to give the accuracy of the prediction model based on the mean squared error (MSE).

     Note: The prediction run on the test dataset must use only the variables identified in the optimized regression model in part D2.

G. Summarize your data analysis by doing the following:
  1. List the packages or libraries you have chosen for Python or R and justify how each item on the list supports the analysis.
  2. Discuss the method used to optimize the model and justification for the approach.
  3. Discuss the verification of assumptions used to create the optimized model.
  4. Provide the regression equation and discuss the coefficient estimates
  5. Discuss the model metrics by addressing each of the following:
     - the R^2 and adjusted R^2 of the training set
     - the comparison of the MSE for the training set to the MSE of the test set
  6. Discuss the results and implications of your prediction analysis.
  7. Recommend a course of action for the real-world organizational situation from part B1 based on your results and implications discussed in part E6.

H. Provide a Panopto video recording that includes *both* a screen share of the presenter demonstrating the functionality of the code used for the analysis and a summary of the programming environment.

   Note: For instructions on how to access and use Panopto, use the "Panopto How-To Videos" web link provided below. To access Panopto's website, navigate to the web link titled "Panopto Access" and then choose to log in using the "WGU" option. If prompted, log in using your WGU student portal credentials, and then it will forward you to Panopto's website.

   To submit your recording, upload it to the Panopto drop box titled "Task 3: Principal Component Analysis – UGN1 | D600". Once the recording has been uploaded and processed in Panopto's system, *retrieve the URL of the recording from Panopto and copy and paste it into the "Links" option. Upload the remaining task requirements using the "Attachments" option.*

I. Acknowledge sources, using in-text citations and references, for content that is quoted, paraphrased, or summarized.

J. Demonstrate professional communication in the content and presentation of your submission.

File Restrictions

File name may contain only letters, numbers, spaces, and these symbols: ! - _ . * ' ( )
File size limit: 200 MB
File types allowed: doc, docx, rtf, xls, xlsx, ppt, pptx, odt, pdf, csv, txt, qt, mov, mpg, avi, mp3, wav, mp4, wma, flv, asf, mpeg, wmv, m4v, svg, tif, tiff, jpeg, jpg, gif, png, zip, rar, tar, 7z

RUBRIC

A:GITLAB REPOSITORY

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
A GitLab repository is not provided.	The subgroup and project are created in GitLab, but 1 or more of the given actions are not completed, or they are completed incorrectly.	The subgroup and project are created in GitLab correctly, and all of the given actions are completed correctly.

B1:PROPOSAL OF QUESTION

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
The submission does not propose 1 question.	The submission proposes 1 question that is not relevant to a real-world organizational situation. Or the proposed question cannot reasonably be answered using linear analysis.	The submission proposes 1 question that is relevant to a real-world organizational situation and can be answered using linear analysis.

B2:DEFINED GOAL

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
The submission does not define 1 goal for data analysis.	The submission defines 1 goal for data analysis, but the goal is not reasonable, is not within	The submission defines 1 reasonable goal for data analysis that is within the scope of the scenario and is represented in

the scope of the scenario, or is not represented in the available data.

the available data.

C1:PCA USE

NOT EVIDENT

The submission does not discuss PCA use.

APPROACHING COMPETENCE

The submission logically explains how PCA can be used to prepare data for regression analysis but does not identify the expected outcomes. Or the submission identifies the expected outcomes but does not explain how PCA can be used to prepare data for regression analysis.

COMPETENT

The submission logically explains how PCA can be used to prepare data for regression analysis and identifies the expected outcomes.

C2:PCA ASSUMPTION

NOT EVIDENT

The submission does not summarize any assumptions of PCA.

APPROACHING COMPETENCE

The submission inaccurately summarizes at least one assumption of PCA.

COMPETENT

The submission accurately summarizes at least one assumption of PCA.

D1:VARIABLE IDENTIFICATION

NOT EVIDENT

The submission does not correctly identify any variables.

APPROACHING COMPETENCE

The submission identifies variables that are relevant to the research question identified in part B1 but not all variables are continuous.

COMPETENT

The submission logically identifies the continuous variables needed to answer the research question identified in part B1.

D2:STANDARDIZED DATA

NOT EVIDENT

A copy of the cleaned dataset is not provided.

APPROACHING COMPETENCE

The submission provides a copy

COMPETENT

The submission appropriately provides a copy of the cleaned

	of the cleaned dataset, but the dataset does not show that the continuous dataset variables identified in part D1 have been standardized.	dataset to show that the continuous dataset variables identified in part D1 have been standardized.

D3:DESCRIPTIVE STATISTICS

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
Screenshots of the descriptive statistics are not provided for the variables identified in part D1.	The submission provides screenshots of the descriptive statistics for some of the variables identified in part D1.	The submission provides screenshots of the descriptive statistics for all of the variables identified in part D1.

E1:MATRIX DETERMINATION

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
The submission does not determine the matrix of any of the principal components.	The submission determines the matrix of some of the principal components.	The submission correctly determines the matrix of all of the principal components.

E2:TOTAL PRINCIPAL COMPONENTS

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
The total number of principal components that should be retained is not identified.	The total number of principal components that should be retained is correctly identified using either the elbow rule or the Kaiser rule but is not supported with a screenshot of the scree plot. Or the total number of principal components is incorrectly identified.	The total number of principal components that should be retained is correctly identified using either the elbow rule or the Kaiser rule and is supported with a screenshot of the scree plot.

E3:VARIANCE

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
The submission does not identify the variance of any of the	The submission identifies the	The submission identifies the variance of *each* of the principal

principal components.	variance of *some* of the principal components from part E2.	components from part E2.

E4:PCA SUMMARY

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
A summary of the results of the PCA is not provided.	The results of the PCA are summarized in an illogical manner.	The results of the PCA are logically summarized.

F1:SPLITTING THE DATA

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
The submission does not provide the training and test dataset file(s).	The submission provides training and test datasets, but the split is not reasonably proportioned.	The submission provides reasonably proportioned training and test datasets.

F2:MODEL OPTIMIZATION

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
The submission does not demonstrate model optimization.	The submission provides an incomplete screenshot of the summary of the optimized model or does not separately identify *all* of the identified model parameters.	The submission suitably demonstrates model optimization by providing a screenshot of the summary of the optimized model or separately identifies *all* of the identified model parameters.

F3:MEAN SQUARED ERROR

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
The mean squared error is not provided.	The mean squared error of a model is provided prior to model optimization.	The mean squared error of the optimized model used on the training set is accurately provided.

F4:MODEL ACCURACY

NOT EVIDENT

The accuracy of the prediction model is not provided.

APPROACHING COMPETENCE

The accuracy of the prediction model is provided, but it is not based on the mean squared error.

COMPETENT

The accuracy of the prediction model is provided based on the mean squared error.

G1:PACKAGES OR LIBRARIES LIST

NOT EVIDENT

The submission does not list the packages or libraries chosen for Python or R.

APPROACHING COMPETENCE

The submission lists the packages or libraries chosen for Python or R but does not justify how 1 or more items on the list support the analysis.

COMPETENT

The submission lists the packages or libraries chosen for Python or R and justifies how each item on the list supports the analysis.

G2:METHOD JUSTIFICATION

NOT EVIDENT

The submission does not discuss the method used to optimize the model.

APPROACHING COMPETENCE

The submission discusses the method used to optimize the model but does not provide justification.

COMPETENT

The submission logically discusses the method used to optimize the model and provides justification.

G3:VERIFICATION OF ASSUMPTIONS

NOT EVIDENT

The submission does not discuss the verification of assumptions used to create the optimized model.

APPROACHING COMPETENCE

The submission discusses the verification of assumptions but does not show a clear connection to model optimization.

COMPETENT

The submission logically discusses the verification of assumptions used to create the optimized model.

G4:EQUATION

NOT EVIDENT

The regression equation is not provided.

APPROACHING COMPETENCE

The regression equation is not

COMPETENT

The regression equation is correctly provided and the coeffi-

	fully provided or the discussion of the coefficient estimates is incorrect.	cient estimates are logically discussed.

G5:MODEL METRICS

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
The submission does not discuss the model metrics.	The discussion of the model metrics does not address *each* of the identified parts.	The submission cogently discusses the model metrics by addressing *each* of the identified parts.

G6:RESULTS AND IMPLICATIONS

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
The submission does not discuss both the results and implications of the prediction analysis.	The submission discusses both the results and implications of the prediction analysis, but the discussion is inadequate.	The submission adequately discusses both the results and implications of the prediction analysis.

G7:COURSE OF ACTION

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
The submission does not recommend a course of action for the real-world organizational situation from part B1.	The submission does not recommend a reasonable course of action for the real-world organizational situation from part B1, or the course of action is not based on the results and implications discussed in part G6, or the course of action recommended is "no response."	The submission recommends a reasonable course of action for the real-world organizational situation from part B1 based on the results and implications discussed in part G6.

H:PANOPTO RECORDING

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
A Panopto video recording is not provided.	The Panopto video recording provided is missing either a screen share of the presenter	The Panopto video recording provided includes *both* a screen share of the presenter demonstrating the functionality of the

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
	demonstrating the functionality of the code used or a discussion commenting on the programming environment. Or either the demonstration or the summary is inaccurate.	code used and a discussion commenting on the programming environment. *Both* the demonstration and the summary are accurate and complete.

I:SOURCES

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
The submission does not include both in-text citations and a reference list for sources that are quoted, paraphrased, or summarized.	The submission includes in-text citations for sources that are quoted, paraphrased, or summarized and a reference list; however, the citations or reference list is incomplete or inaccurate.	The submission includes in-text citations for sources that are properly quoted, paraphrased, or summarized and a reference list that accurately identifies the author, date, title, and source location as available, or the submission states no sources were used.

J:PROFESSIONAL COMMUNICATION

NOT EVIDENT	APPROACHING COMPETENCE	COMPETENT
Content is unstructured, is disjointed, or contains pervasive errors in mechanics, usage, or grammar. Vocabulary or tone is unprofessional or distracts from the topic.	Content is poorly organized, is difficult to follow, or contains errors in mechanics, usage, or grammar that cause confusion. Terminology is misused or ineffective.	Content reflects attention to detail, is organized, and focuses on the main ideas as prescribed in the task or chosen by the candidate. Terminology is pertinent, is used correctly, and effectively conveys the intended meaning. Mechanics, usage, and grammar promote accurate interpretation and understanding.

WEB LINKS

Panopto Access
Sign in using the "WGU" option. If prompted, log in with your WGU student portal credentials, which should forward you to Panopto's website. If you have any problems accessing Panopto, please contact Assessment Services at assessmentservices@wgu.edu. It may take up to two business days to receive your WGU Panopto recording permissions once you have begun the course.

Panopto FAQs

Panopto How-To Videos

WGU GitLab Environment - WGU Community

SUPPORTING DOCUMENTS

D600 Task 3 Dataset 1 Housing Information.csv